

Review Sessions (Power series & Laplace transform)

1. Find a power series solution $x(t) = \sum_{n=0}^{\infty} c_n t^n$ for:

$$x' + x = 0, \quad x(0) = 1 \quad \left(\frac{dx}{dt} + x = 0 \right)$$

Step 1: Assume the power series Let $x(t) = \sum_{n=0}^{\infty} c_n t^n$ (unknown coefficients c_n).

Step 2: Compute x' Differentiate term-by-term:

$$x' = \frac{d}{dt} \left(\sum_{n=0}^{\infty} c_n t^n \right) = \sum_{n=1}^{\infty} n c_n t^{n-1}$$

Let $k = n - 1$, so $n = k + 1$:

$$x' = \sum_{k=0}^{\infty} (k+1) c_{k+1} t^k$$

Step 3: Substitute into $x' + x = 0$ Combine x' and $x(t) = \sum_{k=0}^{\infty} c_k t^k$:

$$\begin{aligned} \sum_{k=0}^{\infty} (k+1) c_{k+1} t^k + \sum_{k=0}^{\infty} c_k t^k &= 0 \\ \sum_{k=0}^{\infty} [(k+1) c_{k+1} + c_k] t^k &= 0 \end{aligned}$$

Step 4: Recurrence Relation (In comparison with the right side, set coefficients to 0) :

$$(k+1) c_{k+1} + c_k = 0 \implies c_{k+1} = -\frac{c_k}{k+1}, \quad k = 0, 1, 2, \dots$$

Step 5: Use the recurrence: - $c_1 = -c_0$, $c_2 = \frac{c_0}{2!}$, $c_3 = -\frac{c_0}{3!}, \dots$, $c_k = \frac{(-1)^k c_0}{k!}$

Step 6: Solution for $x(t)$

$$x(t) = \sum_{k=0}^{\infty} c_k t^k = c_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} t^k$$

Step 7. Let $t = 0 \implies c_0 = x(0) = 1$ (arbitrary constant). Finally

$$x(t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} t^k$$

2. Find a power series solution $x(t) = \sum_{n=0}^{\infty} c_n t^n$ for:

$$x' - 5x = 1 + 2t, \quad x(0) = 1 \quad \left(\frac{dx}{dt} - 5x = 1 + t \right)$$

Step 1: Define the power series

$$x(t) = \sum_{n=0}^{\infty} c_n t^n, \quad \text{where } c_n \text{ are unknown coefficients}$$

Step 2: Compute x' Differentiate term-by-term:

$$x' = \frac{d}{dt} \left(\sum_{n=0}^{\infty} c_n t^n \right) = \sum_{n=1}^{\infty} n c_n t^{n-1}$$

$$\text{Let } k = n - 1 \text{ (so, } n = k + 1) \implies x' = \sum_{k=0}^{\infty} (k + 1) c_{k+1} t^k$$

Step 3: Substitute

$$\sum_{k=0}^{\infty} (k + 1) c_{k+1} t^k - 5 \sum_{k=0}^{\infty} c_k t^k = 1 + 2t$$

$$\sum_{k=0}^{\infty} [(k + 1) c_{k+1} - 5 c_k] t^k = 1 + 2t$$

Step 4: Recurrence Relation

Compare the coefficients in left and right sides, so

$$c_1 - 5c_0 = 1,$$

$$2c_2 - 5c_1 = 2,$$

and other coefficients are 0:

$$(k + 1)c_{k+1} - 5c_k = 0 \implies c_{k+1} = \frac{5c_k}{k + 1}, \quad k = 2, 3, 4, \dots$$

Step 5: Find Coefficients: $c_0 = 1$, $c_1 = 2$ and

$$c_3 = \frac{5c_2}{3},$$

$$c_4 = \frac{5c_3}{3 \times 4} = \frac{5^2 c_2}{3 \times 4},$$

$$c_5 = \frac{5c_4}{3 \times 4 \times 5} = \frac{5^3 c_2}{3 \times 4 \times 5},$$

$$\vdots$$

Step 6: Final Solution: since $c_1 = \frac{2c_2 - 2}{5}$, $c_0 = \frac{2c_2 - 7}{25}$

$$x(t) = \frac{2c_2 - 7}{25} + \frac{2c_2 - 2}{5}t + 2 \times c_2 \left(\frac{t^2}{2} + \frac{5}{3!}t^3 + \frac{5^2}{4!}t^4 + \frac{5^3}{5!}t^5 + \dots \right)$$

Step 7: Find c_2

$$x(0) = \frac{2c_2 - 7}{25} = 1 \implies c_2 = 16$$

$$x(t) = \frac{9}{25} + 6t + 32 \left(\frac{t^2}{2} + \frac{5}{3!}t^3 + \frac{5^2}{4!}t^4 + \frac{5^3}{5!}t^5 + \dots \right)$$

or

$$x(t) = \frac{32}{25}e^{5t} - \frac{7}{25} - \frac{2}{5}t. \quad (\text{Closed form is optional})$$

3. Find a power series solution $x(t) = \sum_{n=0}^{\infty} c_n t^n$ for:

$$x''(t) - 2t x'(t) + x(t) = 0, \quad x(0) = x'(0) = 1$$

Solution

Step 1. Define power series and find its derivatives:

$$x(t) = \sum_{n=0}^{\infty} c_n t^n, \quad x'(t) = \sum_{n=0}^{\infty} (n+1) c_{n+1} t^n, \quad x''(t) = \sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} t^n$$

Step 2. Substitute into the ODE:

$$\sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} t^n - 2 \sum_{n=1}^{\infty} n c_n t^n + \sum_{n=0}^{\infty} c_n t^n = 0$$

Step 3. Compare the left and right sides of the equation (all coefficients are 0)

- For $n = 0$: $2c_2 + c_0 = 0$

- For $n \geq 1$: $c_{n+2} = \frac{(2n-1)c_n}{(n+2)(n+1)}$

Step 4. Compute coefficients:

$$\begin{aligned} c_3 &= \frac{(2 \cdot 1 - 1)c_1}{(1+2)(1+1)} = \frac{c_1}{6}, \\ c_4 &= \frac{(2 \cdot 2 - 1)c_2}{(2+2)(2+1)} = \frac{c_2}{4}, \\ c_5 &= \frac{(2 \cdot 3 - 1)c_3}{(3+2)(3+1)} = \frac{5 \cdot \frac{c_1}{6}}{20} = \frac{c_1}{24}, \\ c_6 &= \frac{(2 \cdot 4 - 1)c_4}{(4+2)(4+1)} = \frac{7 \cdot \frac{c_2}{4}}{30} = \frac{7c_2}{120} \\ &\vdots \end{aligned}$$

Step 5. General Solution ($c_0 = -2c_2$)

$$\begin{aligned} x(t) &= c_0 + c_1 t + c_2 t^2 + c_3 t^3 + c_4 t^4 + c_5 t^5 + \dots \\ &= c_1 \left(t + \frac{t^3}{6} + \frac{t^5}{24} + \dots \right) + c_2 \left(-2 + t^2 + \frac{t^4}{4} + \frac{7t^6}{120} + \dots \right) \end{aligned}$$

Step 6. Apply initial conditions:

$$x(0) = -2c_2 = 1 \implies c_2 = -\frac{1}{2}, \quad \text{and} \quad x'(0) = c_1 = 1$$

Step 7. Final solution

$$x(t) = 1 + t - \frac{1}{2}t^2 + \frac{1}{6}t^3 - \frac{1}{8}t^4 + \frac{1}{24}t^5 - \frac{7}{240}t^6 + \dots$$

4. Use the Laplace transform to solve the initial-value problems

1- $x'' + 2x = 4t, \quad x(0) = 0, \quad x'(0) = 0$

2- $x'' + 4x' + 6x = 1 + e^{-t}, \quad x(0) = 0, \quad x'(0) = 0$

3- $x' + x = e^{-3t} \cos 2t, \quad x(0) = 0$

4- $x'' + x = \sqrt{2} \sin \sqrt{2}t, \quad x(0) = 10, \quad x'(0) = 0$

5- $x'' + 9x = e^t, \quad x(0) = 0, \quad x'(0) = 0$

6- $\frac{dx}{dt} - x = 1, \quad x(0) = 0$

7- $2\frac{dx}{dt} + x = 0, \quad x(0) = -3$

8- $x'' + 5x' + 4x = 0, \quad x(0) = 1, \quad x'(0) = 0$

9- $x'' - 4x' = 6e^{3t} - 3e^{-t}, \quad x(0) = 1, \quad x'(0) = -1$

Solution

(9) $x'' - 4x' = 6e^{3t} - 3e^{-t} \quad (x(0) = 1, x'(0) = -1)$

Step 1. Take Laplace transform

$$\mathcal{L}\{x'' - 4x'\} = \mathcal{L}\{6e^{3t} - 3e^{-t}\}$$

$$s^2X - s - (-1) - 4sX - (-4) = \frac{6}{s-3} - \frac{3}{s+1}$$

$$(s^2 - 4s)X = \frac{6}{s-3} - \frac{3}{s+1} + s - 5$$

$$X(s) = \frac{\frac{6}{s-3} - \frac{3}{s+1} + s - 5}{s(s-4)}$$

Step 2. Simplify and decompose $X(s)$

$$X(s) = \frac{5}{2s} + \frac{11}{10(s-4)} - \frac{2}{s-3} - \frac{3}{5(s+1)}$$

Step 3. Take Inverse Laplace transform

$$x(t) = \frac{5}{2} + \frac{11}{10}e^{4t} - 2e^{3t} - \frac{3}{5}e^{-t}$$